



**CEBU INSTITUTE OF TECHNOLOGY**  
**U N I V E R S I T Y**

# IT342-[Section] System Integration and Architecture

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## **System Design Document (SDD)**

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Project Title: QuickQueue

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Status: Draft

## REVISION HISTORY TABLE

| <b>Versio<br/>n</b> | <b>Date</b> | <b>Author</b>                | <b>Changes Made</b>                 | <b>Status</b> |
|---------------------|-------------|------------------------------|-------------------------------------|---------------|
| 0.1                 | 2/18/2026   | Christian Andrey<br>V. Reyes | Initial draft                       | Draft         |
| 0.2                 | [Date]      | [Your Name]                  | Added API<br>specifications         | Review        |
| 0.3                 | [Date]      | [Your Name]                  | Updated database<br>design          | Review        |
| 0.4                 | [Date]      | [Your Name]                  | Added UI/UX designs                 | Review        |
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| 0.6                 | [Date]      | [Your Name]                  | Final review and<br>corrections     | Final         |
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# EXECUTIVE SUMMARY

## 1.1 Project Overview

QuickQueue is a mobile-based queue management system that allows users to obtain virtual queue numbers for service offices such as clinics, banks, and government agencies. The system reduces physical waiting lines by providing real-time queue updates and notifications.

## 1.2 Objectives

1. Provide a virtual queue system to minimize long waiting times
2. Allow users to monitor queue status remotely
3. Implement a simple and user-friendly mobile application
4. Enable real-time queue updates and notifications
5. Ensure secure access through user authentication

## 1.3 Scope

### Included Features:

- User registration and login
- Service location selection
- Virtual queue number generation
- Queue status monitoring
- Queue notifications

### Excluded Features:

- Online payments
- Appointment scheduling
- Multi-language support
- Advanced analytics
- Offline mode

## 1.0 INTRODUCTION

### 1.1 Purpose

This document defines the system design of QuickQueue, including functional requirements, system architecture, database design, and implementation plan. It serves as a reference for development and integration.

## 2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

### 2.1 Project Overview

**Project Name:** QuickQueue

**Domain:** Queue Management System

**Primary Users:** General public and service office staff

**Problem Statement:** Long waiting lines waste time and cause inconvenience for users.

**Solution:** A mobile application that provides virtual queue numbers and real-time updates.

### 2.2 Core User Journeys

#### Journey 1: First-Time User

1. User installs the app
2. Registers and logs in
3. Selects a service location
4. Takes a queue number
5. Monitors queue status

#### Journey 2: Returning User

1. Logs in
2. Selects service location
3. Takes a new queue number
4. Receives notification when turn is near

## **2.3 Feature List (MoSCoW)**

### **MUST HAVE**

1. User authentication
2. Service location selection
3. Virtual queue number generation
4. Queue status display
5. Queue notification

### **SHOULD HAVE**

1. Estimated waiting time
2. Queue cancellation

### **COULD HAVE**

1. Favorite locations
2. Queue history

### **WON'T HAVE**

1. Payments
2. Appointment booking
3. Admin analytics dashboard

## **2.4 Detailed Feature Specifications**

### **Feature: User Authentication**

- Screens: Register, Login
- Fields: Email, Password
- API: /auth/register, /auth/login

**Feature: Take Queue Number (Transactional Function)**

- Users select service location
- System generates queue number
- API: POST /queue/take

**Feature: Queue Status**

- Displays current number and estimated wait time
- API: GET /queue/status

**Feature: Queue Notification**

- Alerts user when their turn is near

**2.5 Acceptance Criteria****AC-1: Take Queue Number**

Given the user is logged in  
When the user selects a service location  
Then a virtual queue number is generated

**AC-2: Queue Notification**

Given the user is waiting  
When the queue reaches near their number  
Then the user receives a notification

## 3.0 NON-FUNCTIONAL REQUIREMENTS

### 3.1 Performance

- Queue updates  $\leq 2$  seconds
- Support at least 100 concurrent users

### 3.2 Security

- HTTPS communication
- JWT authentication
- Secure password hashing

### 3.3 Compatibility

- Android API Level 24+
- Modern web browsers

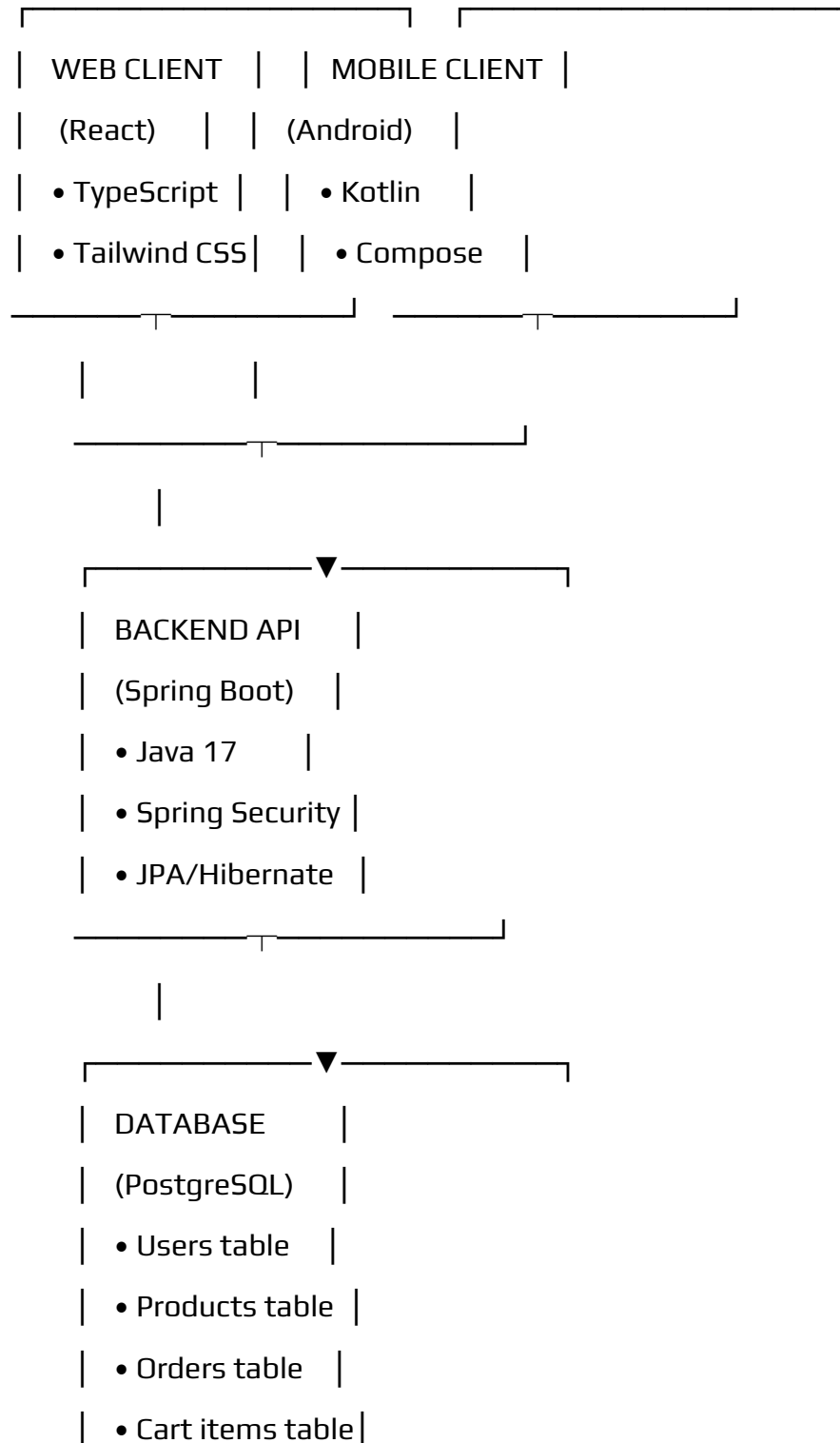
### 3.4 Usability

- Simple interface
- Minimal steps to take queue
- Clear queue information

## 4.0 SYSTEM ARCHITECTURE

### 4.1 Component Diagram

*Note: This should be a component diagram*



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## 4.1 Component Diagram

### Technology Stack:

- **Backend:** Java 17, Spring Boot 3.x, Spring Security (JWT), Spring Data JPA.  
**Database:** PostgreSQL 14+.
- **Web Frontend (Staff Dashboard):** React, TypeScript, Tailwind CSS, Axios.
- **Mobile (Customer App):** Kotlin.
- **Deployment:** Railway/Heroku (Backend), Vercel/Netlify (Web), APK (Mobile).

## 5.0 API CONTRACT & COMMUNICATION

### 5.1 API Standards

- **Base URL:** <https://api.quickqueue.com/api/v1>
- **Format:** JSON for all requests/responses
- **Authentication:** Bearer token (JWT) in Authorization header
- **Response Structure:**

json

```
{  
  "success": boolean,  
  "data": object|null,  
  "error": {  
    "code": string,  
    "message": string,  
    "details": object|null  
  },  
  "timestamp": string  
}
```

### 5.2 Endpoint Specifications

#### Authentication Endpoints

POST /auth/register

Body: {email, password, confirmPassword, fullName?}

Response: {user: {id, email, name}, token, refreshToken}

POST /auth/login

Body: {email, password}

Response: {user: {id, email, name, role}, token, refreshToken}

POST /auth/logout

Headers: Authorization: Bearer {token}

Response: {message: "Logged out successfully"}

## Product Endpoints

GET /products

Query: ?page=1&limit=20&search=keyword&category=electronics

Response: {products: [...], pagination: {page, limit, total, pages}}

GET /products/{id}

Response: {product: {id, name, description, price, stock, imageUrl, category}}

POST /products (Admin only)

Body: {name, description, price, stock, imageUrl, category}

Response: {product: {...}}

PUT /products/{id} (Admin only)

Body: {name?, description?, price?, stock?, imageUrl?, category?}

Response: {product: {...}}

## Cart Endpoints

GET /cart (Authenticated)

Response: {cart: {id, items: [...], total, itemCount}}

POST /cart/items

Body: {productId, quantity}

Response: {message: "Added to cart", cartItem: {...}}

PUT /cart/items/{itemId}

Body: {quantity}

Response: {message: "Cart updated", cartItem: {...}}

DELETE /cart/items/{itemId}

Response: {message: "Removed from cart"}

## Order Endpoints

POST /orders

Body: {shippingAddress: {fullName, address, city, zipCode, country}}

Response: {order: {id, orderNumber, total, status, items: [...], createdAt}}

GET /orders

Response: {orders: [...]}

GET /orders/{id}

Response: {order: {...}}

## 5.3 Error Handling

### HTTP Status Codes

- 200 OK - Successful request
- 201 Created - Resource created
- 400 Bad Request - Invalid input
- 401 Unauthorized - Authentication required/failed
- 403 Forbidden - Insufficient permissions
- 404 Not Found - Resource doesn't exist
- 409 Conflict - Duplicate resource
- 500 Internal Server Error - Server error

### Error Code Examples

json

```
{  
  "success": false,  
  "data": null,  
  "error": {  
    "code": "AUTH-001",  
    "message": "Invalid credentials",  
    "details": "Email or password is incorrect"  
  },  
  "timestamp": "2024-01-28T10:30:00Z"  
}
```

```
{  
  "success": false,  
  "data": null,  
  "error": {
```

```

"code": "VALID-001",
"message": "Validation failed",
"details": {
  "email": "Email is required",
  "password": "Must be at least 8 characters"
},
"timestamp": "2024-01-28T10:30:00Z"
}

```

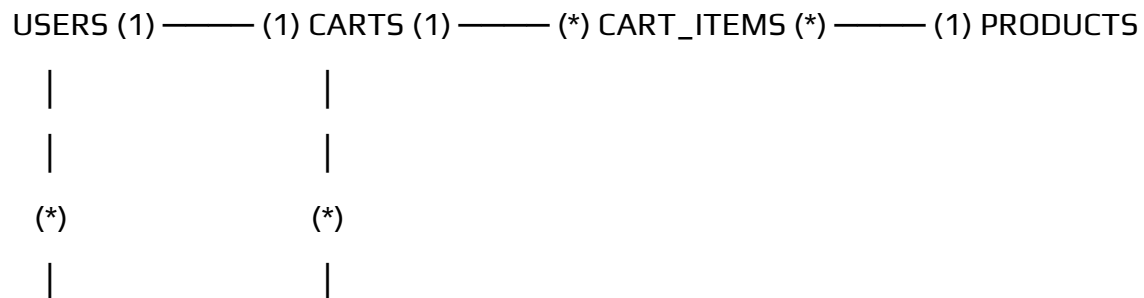
### Common Error Codes

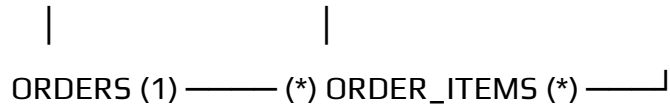
- AUTH-001: Invalid credentials
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- DB-001: Resource not found
- DB-002: Duplicate entry
- BUSINESS-001: Insufficient stock
- SYSTEM-001: Internal server error

## 6.0 DATABASE DESIGN

### 6.1 Entity Relationship Diagram

*Note: This should be an ERD*





### Detailed Relationships:

- **One-to-One:** User  $\leftrightarrow$  Cart (Each user has exactly one cart)
- **One-to-Many:** User  $\rightarrow$  Orders (User can have multiple orders)
- **One-to-Many:** Cart  $\rightarrow$  CartItems (Cart contains multiple items)
- **One-to-Many:** Order  $\rightarrow$  OrderItems (Order contains multiple items)
- **Many-to-One:** CartItems  $\rightarrow$  Product (Items reference products)
- **Many-to-One:** OrderItems  $\rightarrow$  Product (Items reference products)

### Key Tables:

1. **users** - User accounts and authentication
2. **products** - Product catalog information
3. **carts** - Shopping cart per user
4. **cart\_items** - Items in shopping cart
5. **orders** - Customer orders
6. **order\_items** - Items in each order
7. **refresh\_tokens** - JWT refresh tokens

### Table Structure Summary:

- **users:** id, email, password\_hash, full\_name, role, created\_at
- **products:** id, name, description, price, stock, image\_url, category
- **carts:** id, user\_id, created\_at
- **cart\_items:** id, cart\_id, product\_id, quantity
- **orders:** id, order\_number, user\_id, total, status, shipping\_address
- **order\_items:** id, order\_id, product\_id, product\_name, quantity, price

## 7.0 UI/UX DESIGN

## 7.1 Web Application Wireframes

*Note: This should be wireframes from Figma*

## **Homepage (Product Listing)**

Header: [Logo] [Search Bar] [Cart Icon] [User Menu]

Content: Product Grid (3 columns desktop)

Each Product Card: Image, Name, Price, "Add to Cart" button

Footer: Links, Copyright

## **Product Detail Page**

Back Button

Product Image (large)

Product Name and Price

Description

Quantity Selector (1-10)

"Add to Cart" and "Buy Now" buttons

Product Specifications

## **Shopping Cart Page**

Cart Title

List of Cart Items (Image, Name, Quantity, Price, Remove)

Order Summary: Subtotal, Shipping, Tax, Total

"Continue Shopping" and "Proceed to Checkout" buttons

## **Checkout Page**

Shipping Address Form

Order Review (Items, Prices, Totals)

"Place Order" button

Terms and Conditions note

## **Admin Dashboard**

Sidebar Navigation: Dashboard, Products, Orders, Users

Product Management: Add New button, Product list with Edit/Delete

Order Management: Order list with status filters

## **7.2 Mobile Application Wireframes**

*Note: This should be wireframes from Figma*

### **Bottom Navigation**

[ Home] [ Search] [ Cart] [ Profile]

### **Home Screen**

Search Bar

Product Grid (2 columns)

Swipe gestures for quick actions

Pull to refresh

### **Product Detail Screen**

Back arrow

Product image (swipeable gallery)

Product info

Quantity selector

"Add to Cart" fixed bottom button

### **Cart Screen**

Edit mode for quantity updates

Swipe to remove items

Order summary sticky bottom

Checkout button

### **Checkout Flow**

Step indicator: Cart → Shipping → Payment → Confirm

Address form (auto-complete)

Order summary

Place order button

### **Mobile-Specific Features:**

- Touch-optimized buttons (min 44x44px)
- Gesture support (swipe, pull-to-refresh)
- Offline caching for product images
- Bottom navigation for main actions
- Simplified forms for mobile input

### **Design System:**

- **Colors:** Primary (#2563EB), Secondary (#7C3AED), Success (#10B981), Error (#EF4444)
- **Typography:** Inter font family, responsive sizing
- **Spacing:** 8px grid system
- **Components:** Consistent buttons, inputs, cards, modals
- **Responsive:** Mobile-first approach, breakpoints at 640px, 768px, 1024px

## **8.0 PLAN**

### **8.1 Project Timeline**

#### **Phase 1: Planning & Design (Week 1-2)**

Week 1: Requirements & Architecture

Day 1-2: Project setup and documentation

Day 3-4: Complete FRS and NFR

Day 5-7: System architecture design

Week 2: Detailed Design

Day 1-2: API specification

Day 3-4: Database design

Day 5-6: UI/UX wireframes

Day 7: Implementation plan finalization

## **Phase 2: Backend Development (Week 3-4)**

Week 3: Foundation

Day 1: Spring Boot setup with dependencies

Day 2: Database configuration and entities

Day 3: JWT authentication implementation

Day 4: User management endpoints

Day 5: Product CRUD operations

Week 4: Core Features

Day 1: Cart functionality

Day 2: Order management

Day 3: Search and filtering

Day 4: Error handling and validation

Day 5: API documentation and testing

## **Phase 3: Web Application (Week 5-6)**

Week 5: Frontend Foundation

Day 1: React setup with TypeScript

Day 2: Authentication pages (login, register)

Day 3: Product listing page

Day 4: Product detail page

Day 5: Shopping cart implementation

Week 6: Complete Web Features

Day 1: Checkout flow

Day 2: Order history and confirmation

Day 3: Admin dashboard

Day 4: Responsive design polish

Day 5: API integration and testing

#### **Phase 4: Mobile Application (Week 7-8)**

Week 7: Android Foundation

Day 1: Android Studio setup and project structure

Day 2: Authentication screens

Day 3: Product browsing

Day 4: Shopping cart

Day 5: API service layer

Week 8: Complete Mobile App

Day 1: Checkout flow

Day 2: Order management

Day 3: UI polish and animations

Day 4: Testing on emulator/device

Day 5: APK generation and documentation

#### **Phase 5: Integration & Deployment (Week 9-10)**

## Week 9: Integration Testing

Day 1: End-to-end testing across platforms

Day 2: Bug fixes and optimization

Day 3: Security review

Day 4: Performance testing

Day 5: Documentation updates

## Week 10: Deployment

Day 1: Backend deployment (Railway/Heroku)

Day 2: Web app deployment (Vercel/Netlify)

Day 3: Mobile APK distribution

Day 4: Final testing

Day 5: Project submission

### Milestones:

- **M1 (End Week 2):** All design documents complete
- **M2 (End Week 4):** Backend API fully functional
- **M3 (End Week 6):** Web application complete
- **M4 (End Week 8):** Mobile application complete
- **M5 (End Week 10):** Full system deployed and integrated

### Critical Path:

1. Authentication system (Week 3)
2. Product catalog API (Week 3-4)
3. Shopping cart functionality (Week 4)
4. Checkout process (Week 6)
5. Cross-platform testing (Week 9)

### Risk Mitigation:

- Start with simplest working version of each feature

- Test integration points early and often
- Keep backup of working versions
- Focus on core functionality before enhancements